Esercizio 1. Studiare la stabilità del sistema caratterizzato dalla matrice dinamica

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix}$$

$\mathbb{T} = \mathbb{R}$

$$\begin{aligned}
p(s) &= \det(s\,I - A) \\
&= \det\left(\begin{array}{cc|c} s-1 & 0 & 1 \\ -1 & s+1 & -2 \\ \hline 0 & 0 & s+1 \end{array}\right) \\
&= (s-1)\,(s+1)\,(s+1) \\
&= (s-1)\,(s+1)^2 \\
\sigma(A) &= \{1, -1\} \Rightarrow \text{sistema instabile}
\end{aligned}$$

$$p(s) = s^3 + s^2 - s - 1$$

$$-p(s) = -s^3 - s^2 + s + 1$$

$$\begin{array}{c|ll}
3 & 1 & -1 \\
2 & 1 & -1 \\
1 & -1+1=0 & \\
0 & 0 &
\end{array}$$

$$\begin{aligned}
P &= \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix} \\
Q &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
A^\top P + P A + Q &= 0 \\
\begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 0 \\ -1 & 2 & -1 \end{pmatrix} \cdot \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix} + \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} &= 0 \\
\begin{pmatrix} 2\,b + 2\,a + 1 & d & e + 2\,b - a \\ d & 1 - 2\,d & -(2\,e) + 2\,d - b \\ e + 2\,b - a & -(2\,e) + 2\,d - b & -(2\,f) + 4\,e - 2\,c + 1 \end{pmatrix} &= 0 \\
\text{solve}([2\,b + 2\,a + 1, d, e + 2\,b - a, 1 - 2\,d, -(2\,e) + 2\,d - b, -(2\,f) + 4\,e - 2\,c + 1]) & \\
[] &
\end{aligned}$$

$\mathbb{T} = \mathbb{Z}$

$$\sigma(A) = \{1, -1\}$$

$$\text{il sistema non è asintoticamente stabile}$$

λ	μ	ν
1	1	1
-1	2	1

$\Rightarrow$ sistema instabile (un autovalore in modulo pari ad 1 con $\nu < \mu$)

$$\begin{aligned}
\mathrm{Ker}(-1\,I - A) &= \mathrm{Ker}\left(-\mathrm{ident}(3) - \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix}\right) \\
&= \mathrm{Ker}\begin{pmatrix} -2 & 0 & 1 \\ -1 & 0 & -2 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \mathrm{Span}\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
p(z) &= (z-1)\,(z+1)^2 \\
p(1) &= 0 \\
q(s) &= (s-1)^3\,\mathrm{subst}\left(z = \frac{s+1}{s-1}, (z-1)\,(z+1)^2\right) \\
&= 8\,s^2 \\
&\ \text{il sistema non è asintoticamente (esponenzialmente) stabile}
\end{aligned}$$

$$A^{\top} P A - P + Q = 0$$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 0 \\ -1 & 2 & -1 \end{pmatrix} . \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix} . \begin{pmatrix} 1 & 0 & -1 \\ 1 & -1 & 2 \\ 0 & 0 & -1 \end{pmatrix} - \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} d+2\,b+1 & -d-2\,b & -e+2\,d-2\,c+b-a \\ -d-2\,b & 1 & b-2\,d \\ -e+2\,(d+b)-2\,c-b-a & b-2\,d & -(2\,e)+2\,(-e+2\,d-b)+2\,c-2\,b+a+1 \end{pmatrix} = 0$$

$$\text{il sistema non ha soluzione}$$
$$\text{il sistema non è asintoticamente stabile}$$

Esercizio 2. Studiare la stabilità del sistema caratterizzato dalla matrice dinamica

$$A = \begin{pmatrix} -\frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & -\frac{3}{2} & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{3}{2} & 0 \\ 0 & \frac{1}{2} & 0 & -\frac{5}{2} \end{pmatrix}$$

$$\begin{aligned}
p(s) &= \det(s\,I - A) \\
&= \det\left(s\,\mathrm{ident}(4) - \begin{pmatrix} -\frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & -\frac{3}{2} & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & -\frac{3}{2} & 0 \\ 0 & \frac{1}{2} & 0 & -\frac{5}{2} \end{pmatrix}\right)
\end{aligned}$$

$$
\begin{aligned}
&= \det\begin{pmatrix} s+\frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & s+\frac{3}{2} & 0 & \frac{1}{2} \\ -(\frac{1}{2}) & 0 & s+\frac{3}{2} & 0 \\ 0 & -(\frac{1}{2}) & 0 & s+\frac{5}{2} \end{pmatrix} \\
&= \left(s+\frac{1}{2}\right)\det\begin{pmatrix} s+\frac{3}{2} & 0 & \frac{1}{2} \\ 0 & s+\frac{3}{2} & 0 \\ -(\frac{1}{2}) & 0 & s+\frac{5}{2} \end{pmatrix} - \left(\frac{1}{2}\right)\det\begin{pmatrix} 0 & \frac{1}{2} & 0 \\ s+\frac{3}{2} & 0 & \frac{1}{2} \\ -(\frac{1}{2}) & 0 & s+\frac{5}{2} \end{pmatrix} \\
&= \left(s+\frac{1}{2}\right)\left(s+\frac{3}{2}\right)\det\begin{pmatrix} s+\frac{3}{2} & \frac{1}{2} \\ -(\frac{1}{2}) & s+\frac{5}{2} \end{pmatrix} - \frac{1}{2}\frac{1}{2}\det\begin{pmatrix} s+\frac{3}{2} & \frac{1}{2} \\ -(\frac{1}{2}) & s+\frac{5}{2} \end{pmatrix} \\
&= \left(s+\frac{1}{2}\right)\left(s+\frac{3}{2}\right)\left(\left(s+\frac{3}{2}\right)\left(s+\frac{5}{2}\right)+\frac{1}{4}\right) - \frac{1}{4}\left(\left(s+\frac{3}{2}\right)\left(s+\frac{5}{2}\right)+\frac{1}{4}\right) \\
&= \left(\left(s+\frac{3}{2}\right)\left(s+\frac{5}{2}\right)+\frac{1}{4}\right)\left(\left(s+\frac{1}{2}\right)\left(s+\frac{3}{2}\right)-\frac{1}{4}\right) \\
&= (s^2+4\,s+4)\left(s^2+2\,s+\frac{1}{2}\right)
\end{aligned}
$$

Per $\mathbb{T}=\mathbb{R}$, applicando il criterio di Cartesio due volte concludo che il sistema è asintoticamente stabile

$$
\begin{aligned}
p_1(z) &= z^2+4\,z+4 \\
p_1(1) &\neq 0 \\
q_1(s) &= (s-1)^2\,\mathrm{subst}\left(z=\frac{s+1}{s-1}, z^2+4\,z+4\right) \\
&= 9\,s^2-6\,s+1
\end{aligned}
$$

$q_1(s)$ non è Hurwitz $\Rightarrow$ $p_1(z)$ non è Schur $\Rightarrow$ $p(z)$ non è Schur (teorema fondmanetale dell'algebra)

Esercizio 3. Determinare i punti di equilibrio del sistema

$$
\begin{aligned}
x_1(t+1) &= \frac{1}{2}\,x_1(t)+x_1(t)\,x_2(t)+x_3(t) \\
x_2(t+1) &= x_1^2(t)-\frac{2}{3}\,x_2(t)+2\,x_3(t) \\
x_3(t+1) &= \frac{1}{3}\,x_3(t)
\end{aligned}
$$

e valutarne la stabilità locale per il sistema nonlineare.

$$
\begin{aligned}
\frac{1}{2}\,x_1+x_1\,x_2+x_3 &= x_1 \\
x_1^2-\frac{2}{3}\,x_2+2\,x_3 &= x_2 \\
\frac{1}{3}\,x_3 &= x_3
\end{aligned}
$$

$$
\begin{aligned}
x_3 &= 0 \\
x_1^2-\frac{5}{3}\,x_2 &= 0
\end{aligned}
$$

$$
\begin{aligned}
-\frac{1}{2}\,x_1 + x_1\,x_2 &= 0\\[2ex]
x_2 &= \frac{3}{5}\,x_1^2\\
-\frac{1}{2}\,x_1 + \frac{3}{5}\,x_1^3 &= 0\\
-\frac{1}{10}\,x_1\,(5 - 6\,x_1^2) &= 0\\[2ex]
x_1 &= 0\\
x_2 &= 0\\
x_3 &= 0\\[2ex]
x_1 &= \sqrt{\frac{5}{6}}\\
x_2 &= \frac{1}{2}\\
x_3 &= 0\\[2ex]
x_1 &= -\sqrt{\frac{5}{6}}\\
x_2 &= \frac{1}{2}\\
x_3 &= 0\\[2ex]
\nabla_x f(x) &= \begin{pmatrix} \frac{1}{2}+x_2 & x_1 & 1\\ 2\,x_1 & -\frac{2}{3} & 2\\ 0 & 0 & \frac{1}{3}\end{pmatrix}
\end{aligned}
$$

$x_{e,1} = \begin{pmatrix} 0\\ 0\\ 0\end{pmatrix} \Rightarrow A = \begin{pmatrix} \frac{1}{2} & 0 & 1\\ 0 & -\frac{2}{3} & 2\\ 0 & 0 & \frac{1}{3}\end{pmatrix} \Rightarrow \sigma(A) = \left\{\frac{1}{2}, -\frac{2}{3}, \frac{1}{3}\right\} \Rightarrow$ il punto di equilibrio $x_{e,1}$ è ES per il sistema-nonlineare

$x_{e,2} = \begin{pmatrix} \sqrt{\frac{5}{6}}\\ \frac{1}{2}\\ 0\end{pmatrix} \Rightarrow A = \begin{pmatrix} 1 & \sqrt{\frac{5}{6}} & 1\\ 2\sqrt{\frac{5}{6}} & -\frac{2}{3} & 2\\ 0 & 0 & \frac{1}{3}\end{pmatrix} \Rightarrow p(z) = \det\left(\begin{array}{cc|c} z-1 & -\left(\frac{\sqrt{5}}{\sqrt{6}}\right) & -1\\ -\left(\frac{2\sqrt{5}}{\sqrt{6}}\right) & z+\frac{2}{3} & -2\\ \hline 0 & 0 & z-\frac{1}{3}\end{array}\right)$

$p(z) = \left(z-\frac{1}{3}\right)\left((z-1)\left(z+\frac{2}{3}\right)-\frac{5}{3}\right) = \left(z-\frac{1}{3}\right)\left(z^2-\frac{z}{3}-\frac{7}{3}\right)$

$p_2(z) = z^2-\frac{z}{3}-\frac{7}{3} \Rightarrow q_2(s) = (s-1)^2\,\mathrm{subst}\left(z=\frac{s+1}{s-1}, z^2-\frac{z}{3}-\frac{7}{3}\right) = -\left(\frac{5\,s^2-20\,s+3}{3}\right) = -\frac{5}{3}\,s^2+\frac{20}{3}\,s-1$

il punto di equilibrio $x_{e,2}$ non è ES per il sistema nonlineare

$\mathrm{solve}\left(z^2-\frac{z}{3}-\frac{7}{3}, z\right) = \left[z = -\left(\frac{\sqrt{85}-1}{6}\right), z = \frac{\sqrt{85}+1}{6}\right] \Rightarrow$ un autovalore in modulo maggiore di 1

il punto di equilibrio $x_{e,2}$ è instabile per il sistema nonlineare

$$x_{e,3} = \begin{pmatrix} -\sqrt{\frac{5}{6}} \\ \frac{1}{2} \\ 0 \end{pmatrix} \Rightarrow A = \begin{pmatrix} 1 & -\sqrt{\frac{5}{6}} & 1 \\ -2\sqrt{\frac{5}{6}} & -\frac{2}{3} & 2 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} \Rightarrow (z) = \left(z - \frac{1}{3}\right)\left((z-1)\left(z+\frac{2}{3}\right) - \frac{5}{3}\right) = \left(z - \frac{1}{3}\right)\left(z^2 - \frac{z}{3} - \frac{7}{3}\right)$$

il punto di equilibrio $x_{e,3}$ è instabile per il sistema nonlineare

sistema linearizzato		**sistema nonlineare**
esponenzialmente (asintoticamente) stabile	$\Rightarrow$	esponenzialmente stabile
stabile	$\Rightarrow$	non posso concludere nulla (non è esp. stabile) potrebbe essere - asintoticamente stabile - stabile semplicemente - instabile
instabile	$\Rightarrow$	instabile

Esercizio 4. Si determini una realizzazione della seguente f.d.t

$$W(s) = \frac{s^3 + 2s^2 + 2s + 1}{s^4 + 2s^3 + 3s^2 + 4s + 1}$$

$$\begin{aligned} D &= \lim_{s\to\infty} W(s) \\ &= 0 \end{aligned}$$

$$\begin{aligned} A_c &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & -4 & -3 & -2 \end{pmatrix} \\ B_c &= \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\ C_c &= \begin{pmatrix} 1 & 2 & 2 & 1 \end{pmatrix} \\ D_c &= 0 \end{aligned}$$

$$\begin{aligned} A_o &= \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -2 \end{pmatrix} \\ B_o &= \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix} \\ C_o &= \begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix} \\ D_o &= 0 \end{aligned}$$

Esercizio 5. Calcolare la risposta in uscita al sistema dinamico per $\mathbb{T}=\mathbb{Z}$ caratterizzato dalle seguenti matrici

$$
\begin{aligned}
A &= \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \\
B &= \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\
C &= \begin{pmatrix} 1 & 0 \end{pmatrix} \\
D &= -1
\end{aligned}
$$

$$
\begin{aligned}
u(t) &= (-1)^t \delta_{-1}(t) + \delta_0(t-1) \\
x_0 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}
\end{aligned}
$$

ed evidenziare risposta pseudo-permanetente e componente pseudo-transitoria.

$$
\begin{aligned}
z\,I - A &= z\,\mathrm{ident}(2) - \begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix} \\
&= \begin{pmatrix} z+1 & -1 \\ 0 & z+1 \end{pmatrix} \\
(z\,I - A)^{-1} &= \frac{1}{(z+1)^2} \begin{pmatrix} z+1 & 1 \\ 0 & z+1 \end{pmatrix} \\
&= \begin{pmatrix} \frac{1}{z+1} & \frac{1}{(z+1)^2} \\ 0 & \frac{1}{z+1} \end{pmatrix} \\
z\,(z\,I - A)^{-1} &= \begin{pmatrix} \frac{z}{z+1} & \frac{z}{(z+1)^2} \\ 0 & \frac{z}{z+1} \end{pmatrix} \\
A^t &= \begin{pmatrix} (-1)^t & (-1)^{t-1}\binom{t}{1} \\ 0 & (-1)^t \end{pmatrix}
\end{aligned}
$$

$$
\begin{aligned}
H(z) &= z\,C\,(z\,I - A)^{-1} \\
&= z \begin{pmatrix} 1 & 0 \end{pmatrix} . \begin{pmatrix} \frac{1}{z+1} & \frac{1}{(z+1)^2} \\ 0 & \frac{1}{z+1} \end{pmatrix} \\
&= \begin{pmatrix} \frac{z}{z+1} & \frac{z}{(z+1)^2} \end{pmatrix} \\
y_l(z) &= H(z)\,x_0 \\
&= \begin{pmatrix} \frac{z}{z+1} & \frac{z}{(z+1)^2} \end{pmatrix} . \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
&= \frac{z}{z+1} \\
y_l(t) &= (-1)^t
\end{aligned}
$$

$$
\begin{aligned}
W(z) &= C\,(z\,I - A)^{-1}\,B + D \\
&= \begin{pmatrix} 1 & 0 \end{pmatrix} . \begin{pmatrix} \frac{1}{z+1} & \frac{1}{(z+1)^2} \\ 0 & \frac{1}{z+1} \end{pmatrix} . \begin{pmatrix} 0 \\ 1 \end{pmatrix} - 1 \\
&= \frac{1}{(z+1)^2} - 1
\end{aligned}
$$

$$
\begin{aligned}
u(z) &= \frac{z}{z+1}+\frac{1}{z} \\
\\
y_f(z) &= \left(\frac{1}{(z+1)^2}-1\right)\left(\frac{z}{z+1}+\frac{1}{z}\right) \\
&= \frac{z}{(z+1)^3}+\frac{1}{z\,(z+1)^2}-\frac{z}{z+1}-\frac{1}{z} \\
\\
\frac{1}{z\,(z+1)^2} &= \frac{z}{z^2\,(z+1)^2} \\
&= z\left(\frac{1}{z^2\,(z+1)^2}\right) \\
&= z\left(\frac{A}{z}+\frac{B}{z^2}+\frac{C}{z+1}+\frac{D}{(z+1)^2}\right) \\
B &= \frac{1}{(0+1)^2}=1 \\
D &= \frac{1}{(-1)^2}=1 \\
\lim_{z\to\infty}\frac{z}{z^2\,(z+1)^2} &= \lim_{z\to\infty} z\left(\frac{A}{z}+\frac{B}{z^2}+\frac{C}{z+1}+\frac{D}{(z+1)^2}\right) \\
0 &= A+C \\
\frac{A}{1}+\frac{B}{1^2}+\frac{C}{1+1}+\frac{D}{(1+1)^2} &= \frac{1}{1^2\,(1+1)^2} \\
A+B+\frac{1}{2}C+\frac{1}{4}D &= \frac{1}{4} \\
-C+1+\frac{1}{2}C+\frac{1}{4} &= \frac{1}{4} \\
-\frac{1}{2}C &= -1 \\
C &= 2 \\
A &= -2 \\
\\
\frac{1}{z\,(z+1)^2} &= \frac{z}{(z+1)^2}+\frac{2\,z}{z+1}+\frac{1}{z}-2 \\
y_f(z) &= \frac{z}{(z+1)^3}+\frac{1}{z\,(z+1)^2}-\frac{z}{z+1}-\frac{1}{z} \\
&= \frac{z}{(z+1)^3}+\frac{z}{(z+1)^2}+\frac{z}{z+1}-2 \\
y_f(t) &= (-1)^{t-2}\binom{t}{2}+(-1)^{t-1}\binom{t}{1}+(-1)^t-2\,\delta_0(t) \\
\\
y(t) &= y_l(t)+y_f(t) \\
&= (-1)^t+(-1)^{t-2}\binom{t}{2}+(-1)^{t-1}\binom{t}{1}+(-1)^t-2\,\delta_0(t) \\
&= \underbrace{(-1)^{t-1}\binom{t}{1}+2\,(-1)^t}_{\text{comp. pseudo}-\text{transitoria}}+\underbrace{(-1)^{t-2}\binom{t}{2}-2\,\delta_0(t)}_{\text{sol. pseudo}-\text{permanente}}
\end{aligned}
$$